

Smart Parking Space Rental & License Validator

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1. Architectural Choice

We chose Layered Architecture for the Smart Parking Space Rental & License Validator system. The presentation, business logic, and data/external-service responsibilities are separated into distinct layers.

2. Reason 1 – Separate user responsibilities

The system has two user roles: the Vehicle Owner and Parking Warden. The Vehicle Owner Portal supports parking-space search and rental, while the Warden Console supports license validation, inspection, and parking-session monitoring. This keeps user-interface responsibilities separate from core business rules.

3. Reason 2 – Controlled business processing

The business layer separates Space Rental Service, License Validation Service, Session & Rule Service, and the Access Control Adapter. This makes rental, validation, session rules, and entry/exit integration distinct responsibilities that can communicate through defined interfaces.

4. Security Advantage

The presentation components do not directly access the Parking Data Store. Requests pass through business services, where authorization and validation rules can be applied. License verification is also routed through a dedicated validation service and external registry interface.

5. Performance Benefit

Parking availability and active-session information can be retrieved through dedicated business services and optimized data queries. The separation reduces unnecessary processing in the presentation layer and allows the data store to be tuned for frequent parking lookups.

6. Component & Interface Summary

Components: Vehicle Owner Portal, Warden Console, Space Rental Service, License Validation Service, Session & Rule Service, Access Control Adapter, Parking Data Store, Payment Gateway, License Registry API. Interfaces: booking request, validation request, rental interface, validation interface, license check, session rules, access decision, payment request, data access, session records, and license lookup.

Design basis: Original design prepared for the assigned Smart Parking scenario and Lab 3 component-modelling requirements.